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Improving Space Complexity

↳ For calculating the score of ^{current} one column we only need the previous column.

↳ We can throw away that previous column once we are done using it.

→ Middle Edge: An edge in an optimal alignment path starting at the middle node.

→ Recursive LinearSpaceAlignment Algo $O(nm)$

LinearSpaceAlignment (top, bottom, left, right)

if left == right

return alignment formed by bottom-top edges "↓"

middle = $\lfloor (left + right) / 2 \rfloor$

midNode = MiddleNode (top, bottom, left, right)

midEdge = MiddleEdge (top, bottom, left, right)

LinearSpaceAl — (top, midNode, left, middle)

output midEdge

if midEdge == "→" || midEdge == "↘"

middle = middle + 1

if midEdge == "↓" || midEdge == "↙"

midNode = midNode + 1

Linear — (midNode, bottom, middle, right)

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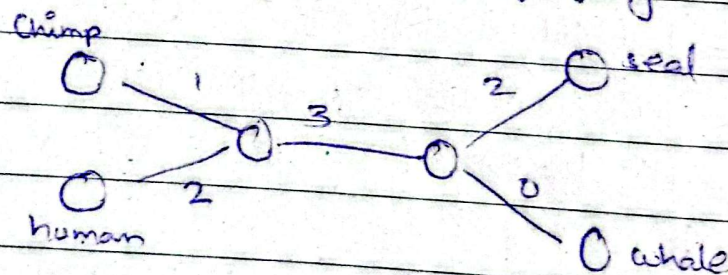
▷ Additive Phylogeny In Action

• Distance Matrix

↳ A matrix which stores no. of differing symbols b/w multiple alignments

Species	Alignment	Distance Matrix			
		Chimp	Human	Seal	Whale
Chimp	ACGTAGG CCT	0	3	6	4
Human	ATGGTAAGACT	3	0	7	5
Seal	TGAGAGG GAC	6	7	0	2
Whale	TGAAAGCAT	4	5	2	0

→ The distance matrix can be used to make an evolutionary tree (Phylogeny Tree)



→ Additive matrix is a distance matrix for which there exist an unrooted tree fitting it

→ Simple Tree — tree with no nodes of degree 2

→ There is a unique simple tree fitting additive matrix

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→ Limb Length :

Limb length (i) is equal to the minimum value of $(D_{i,k} + D_{i,j} + D_{j,k}) / 2$ over all leaves j & k .

↳ Limb Length (chimp):

$$= (D_{\text{chimp, human}} + D_{\text{chimp, seal}} - D_{\text{human, seal}}) / 2$$

$$= (3 + 6 - 7) / 2 = 1$$

$$= (D_{\text{chimp, human}} + D_{\text{chimp, whale}} - D_{\text{human, whale}}) / 2$$

$$= (3 + 4 - 5) / 2 = 1 \quad \checkmark$$

$$= (D_{\text{chimp, seal}} + D_{\text{chimp, whale}} - D_{\text{seal, whale}}) / 2$$

$$= (6 + 4 - 2) / 2 = 4$$

→ Algorithm

- ① Pick an arbitrary leaf j .
- ② Compute $\text{LimbLength}(j)$.
- ③ Subtract $\text{LimbLen}(j)$ from the row & col of j to produce D^{bald} in which j is a bald limb (length 0).
- ④ Remove the j^{th} row & col to form D^{trim} .
- ⑤ Construct Tree (D^{trim}).
- ⑥ Identify the point in Tree (D^{trim}) where leaf j should be attached.

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6.1 First find minimum value of $(D^{bald}_{ij} + D^{bald}_{jk} - D^{bald}_{ik}) / 2$ over all leaves i & k

6.2 The attachment point for j is found on the path b/w leaves i & k at distance $D^{bald}_{i,j}$ from i .

Attach j by an edge of length $LimbLength(j)$ in order to form Tree (D)

Q	i	j	k	l
i	0			
j	13	0		
k	21	12	0	
l	22	13	13	0

① Limb length (l)

$$= (D_{l,i} + D_{l,j} - D_{i,j}) / 2 = (22 + 13 - 13) / 2 = 11$$

$$= (D_{l,i} + D_{l,k} - D_{i,k}) / 2 = (22 + 13 - 21) / 2 = 7$$

$$= (D_{l,j} + D_{l,k} - D_{j,k}) / 2 = (13 + 13 - 12) / 2$$

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D^{old}	i	j	k	l
i	0			
j	13	0		
k	21	12	0	
l	15	6	6	0

$$D^{\text{old}}(i, k) = D^{\text{old}}(l, i) + D^{\text{old}}(l, k)$$

$$21 = 15 + 6 \quad \checkmark$$

Attachment distance $\{21 \geq D^{\text{old}}(i, l) = 15\}$

D^{form}	i	j	k
i	0		
j	13	0	
k	21	12	0

② Unb length k :

$$= (D_{i,k} + D_{j,k} - D_{i,j}) / 2 = (21 + 12 - 13) / 2$$

$$= 10 \quad \checkmark$$

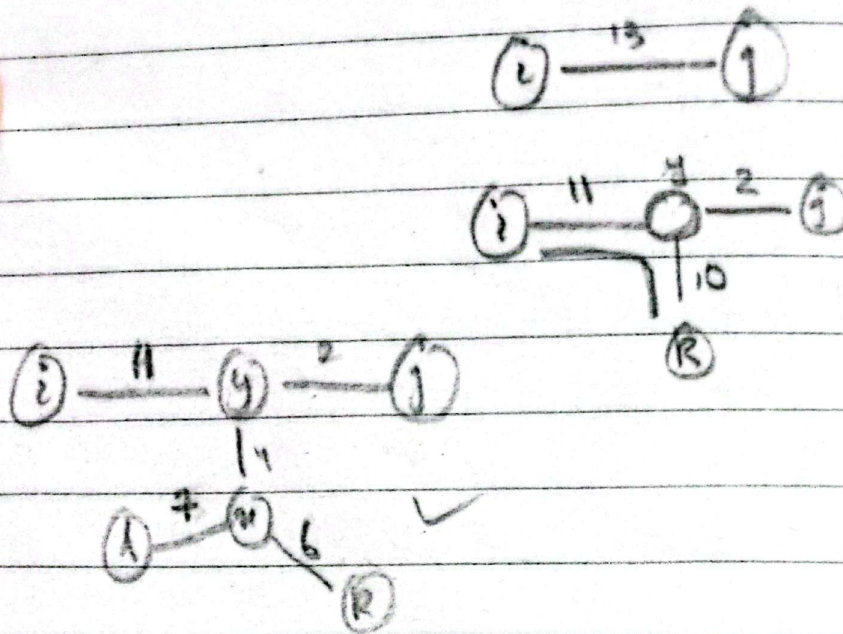
D^{old}	i	j	k
i	0		
j	13	0	
k	11	2	0

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$$D(i, j) = D(k, i) + D(k, j) = 13$$

$$\text{Attaching distance} = D(k, i) = 11$$



Q:

	x	y	z	t
x	0	13	21	22
y	13	0	12	13
z	21	12	0	13
t	22	13	13	0

Limb Length (t)

$$(D_{t,x} + D_{t,y} - D_{x,y}) / 2 = (22 + 13 - 13) / 2 =$$

$$D(t, z + D_{t,z} - D_{x,z}) / 2 = (22 + 13 - 21) / 2 =$$

$$D(t, y + D_{t,y} - D_{y,z}) / 2 =$$

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	A	B	C	D
A	0			
B	5	0		
C	9	10	0	
D	9	10	8	0

• Limb length D

$$= (D_{D,A} + D_{D,B} - D_{A,B}) = (9 + 10 - 5)/2 = 7$$

$$= (D_{D,A} + D_{D,C} - D_{A,C}) = (9 + 8 - 9)/2 = (4) \checkmark$$

$$= (D_{D,B} + D_{D,C} - D_{B,C}) = (10 + 8 - 10)/2 = 4$$

Limb length (D) = 4

D'	A	B	C	D
A	0			
B	5	0		
C	9	10	0	
D	5	6	4	0

$$D'(A,C) = D'(D,A) + D'(D,C) - 9 \checkmark$$

~~D lies between~~ Distance $x = D(A,x) + 5$
A to x

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Distan	A	B	C
A	0		
B	5	0	
C	9	10	0

• Ind leg (C)

$$\rightarrow (D_{C,B} + D_{C,A} - D_{B,A}) = (10 + 9 - 5) = 14$$

$$= 7$$

D'	A	B	C
A	0		
B	5	0	
C	2	3	0

$$7 - 5 = 2$$

Attaching distan $y = 2$

